

Phrase 15

Table 1. ArtiFact + COCO2017 Zero-Shot Evaluation Results

Metric	Value
Accuracy	0.5762
Precision	0.5468
Recall	0.8854
F1-Score	0.6761
Specificity	0.2676
AUC-ROC	0.5557

Table 2. Confusion Matrix

	Predicted FAKE	Predicted REAL
Actual FAKE	4,418 (TP)	572 (FN)
Actual REAL	3,662 (FP)	1,338 (TN)

Table 3. Zero-Shot Evaluation Dataset

Dataset Component	Description
Real Images	COCO 2017 val2017 (5,000 images)
Fake Images	ArtiFact (4,990 images, 33 AI generators)

Table 4. Evaluation Interpretation

Observation	Interpretation
Accuracy	The model achieves 57.62% overall accuracy under zero-shot evaluation.
Precision	A precision of 54.68% indicates moderate reliability when predicting fake images.

Recall	The high recall (88.54%) shows that most fake images are successfully detected.
F1-Score	The F1-score of 67.61% reflects moderate overall detection performance.
Specificity	The low specificity (26.76%) indicates that many real images are incorrectly classified as fake.
AUC-ROC	An AUC-ROC of 0.5557 suggests limited discrimination capability on unseen datasets.
Key Finding	The noticeable performance drop compared with CIFAKE demonstrates limited cross-dataset generalization. The model detects fake images well but struggles to distinguish real images from an unseen distribution, highlighting the domain shift challenge in zero-shot deepfake detection.



